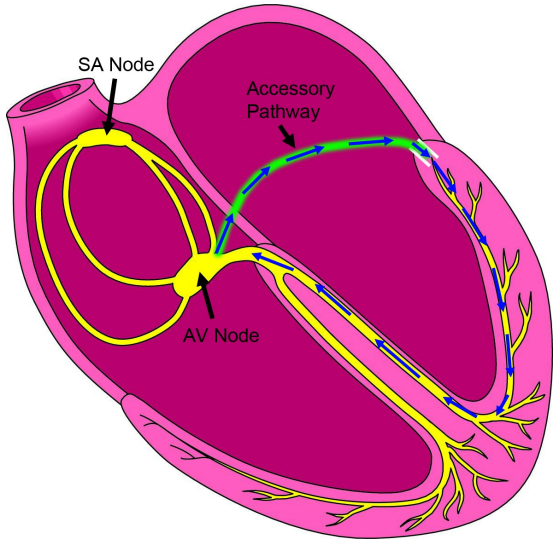
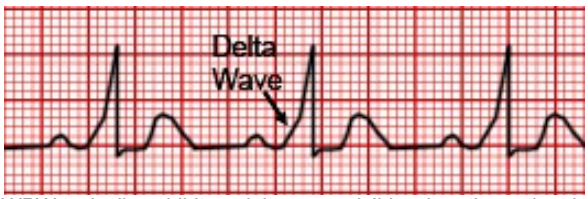
Wolff Parkinson White Syndrome *Prodigy Anesthesia*

Wolf Parkinson White syndrome is a defect in the cardiac conduction system in which an extra conduction pathway called the bundle of Kent exists that allows the transmission of electrical impulses via an abnormal circuit.

Normally, electrical impulses travel from the sinoatrial (SA) node to the atrioventricular (AV) node, which acts as a gatekeeper that limits the number of impulses that can travel to the ventricles via the bundle of His. The abnormal pathway, however, does not possess this rate-controlling characteristic. If an impulse travels through the abnormal bypass tract while the AV nodal tissue is refractory, it can travel via the Purkinje fibers, back up the bundle of His, and back through the abnormal pathway. Via this pathway, rates can be extremely rapid, typically 150-250 beats per minute. The event that typically triggers an episode of WPW is a premature electrical impulse.



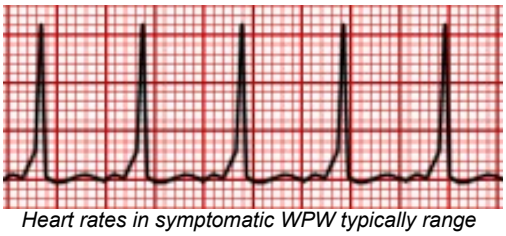
The electrical circuit can travel either retrograde, as described above or anterograde. In an anterograde circuit, the accessory pathway is refractory during the premature triggering impulse and the electrical current passes through the Purkinje fibers, reaches the accessory pathway, and travels in a loop in the opposite direction.



WPW typically exhibits a delta wave visible when the patient is asymptomatic.

WPW is typically diagnosed by the presence of a delta wave on the ECG when the patient is asymptomatic (has a normal heart rate). The PR interval is normal shortened (<0.12s). The rhythm is often regular, although atrial fibrillation can coexist with WPW. The QT interval is normally > 0.10s and the QRS appears slurred by the delta wave.

Although propofol, opioids, and benzodiazepines have little, if any, effect on the conduction system, volatile anesthetics increase anterograde refractoriness in both normal and accessory pathways and increase the ability of an extrasystole to trigger tachycardia.



Heart rates in symptomatic WPW typically range from 150-250 bpm.

Agents that increase sympathetic tone, such as ketamine, glycopyrrolate, atropine, pancuronium, and sympathomimetics should be used with caution or avoided if possible. Light anesthesia, hypercapnia, and acidosis can trigger the sympathetic system and increase the likelihood that an extrasystole will result in tachycardia. Be certain that the patient is under deep anesthesia prior to intubation and deep extubation, if possible, may reduce the likelihood of tachyarrhythmias on emergence. Avoid verapamil or digoxin (which could enhance anterograde conduction through an accessory pathway) in the treatment of any arising dysrhythmia, and have beta blockers, adenosine, and/or amiodarone available for treatment of tachydysrhythmias.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 370-371.
Barash PG, Cullen BF, Stoelting RK. Clinical Anesthesia. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 1545.

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